Exam

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In [1]:
```
import pandas as pd
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from tensorflow.keras.optimizers import Adam
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import MinMaxScaler
import matplotlib.pyplot as plt
```

```
2025-04-21 11:03:36.128648: I tensorflow/core/platform/cpu_feature_guard.cc:182] This TensorFlow binary is optimized to
use available CPU instructions in performance-critical operations.
To enable the following instructions: AVX2 FMA, in other operations, rebuild TensorFlow with the appropriate compiler f
lags.
```

In [2]:
```
df = pd.read_csv("Synthetic_app_data.csv")
print(df.head())
```

```
   Average Response Time (s)  Number of Features  Number of Bugs Reported  \
0                   1.436350                  49                       16
1                   2.876786                  36                       12
2                   2.329985                  34                        0
3                   1.996646                  39                        1
4                   0.890047                  44                        8

   Training Hours Provided  User Satisfaction Score
0                 3.970126                89.980978
1                 3.100364                56.732146
2                 2.667305               121.502856
3                 4.469463               124.535638
4                 3.942986               129.077397
```

In [3]:
```
print(df.columns.tolist())
```

```
['Average Response Time (s)', 'Number of Features', 'Number of Bugs Reported', 'Training Hours Provided', 'User Satisfa
ction Score']
```

Q1: Split Dataset 70/30

In [4]:
```
features = ['Average Response Time (s)', 'Number of Features', 'Number of Bugs Reported', 'User Satisfaction Score']
target = 'Training Hours Provided'
X = df[features]
y = df[target]

# Split into 70% training and 30% Validation
X_train, X_val, y_train, y_val = train_test_split(X, y, test_size=0.3, random_state=42)
```

Q2: Use mean_squared_error loss function

In [5]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

In [6]:
```
model = Sequential([
    Dense(64, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(32, activation='relu'),
    Dense(1)  #output layer
])
model.compile(optimizer=Adam(), loss='mean_squared_error', metrics=['mae'])

history = model.fit(X_train, y_train, epochs=100, validation_data=(X_val, y_val), verbose=1)
```

```
Epoch 1/100
7/7 [==============================] - 1s 25ms/step - loss: 5.7537 - mae: 1.9153 - val_loss: 3.0664 - val_mae: 1.4571
Epoch 2/100
7/7 [==============================] - 0s 4ms/step - loss: 3.3683 - mae: 1.4818 - val_loss: 2.4987 - val_mae: 1.3393
Epoch 3/100
7/7 [==============================] - 0s 4ms/step - loss: 3.0802 - mae: 1.4561 - val_loss: 2.1215 - val_mae: 1.2408
Epoch 4/100
7/7 [==============================] - 0s 4ms/step - loss: 2.6451 - mae: 1.3496 - val_loss: 2.3344 - val_mae: 1.2804
Epoch 5/100
7/7 [==============================] - 0s 4ms/step - loss: 2.4914 - mae: 1.3262 - val_loss: 1.9022 - val_mae: 1.1950
Epoch 6/100
7/7 [==============================] - 0s 4ms/step - loss: 2.2743 - mae: 1.2890 - val_loss: 1.9692 - val_mae: 1.2005
Epoch 7/100
7/7 [==============================] - 0s 4ms/step - loss: 2.2521 - mae: 1.2787 - val_loss: 1.8308 - val_mae: 1.1492
Epoch 8/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1080 - mae: 1.2436 - val_loss: 1.8193 - val_mae: 1.1490
Epoch 9/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0279 - mae: 1.2273 - val_loss: 1.7880 - val_mae: 1.1536
Epoch 10/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0608 - mae: 1.2355 - val_loss: 1.7988 - val_mae: 1.1505
Epoch 11/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0374 - mae: 1.2426 - val_loss: 1.7923 - val_mae: 1.1524
Epoch 12/100
7/7 [==============================] - 0s 4ms/step - loss: 2.2388 - mae: 1.2889 - val_loss: 1.7836 - val_mae: 1.1593
Epoch 13/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9912 - mae: 1.2165 - val_loss: 1.7880 - val_mae: 1.1627
Epoch 14/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9296 - mae: 1.2050 - val_loss: 1.7942 - val_mae: 1.1608
Epoch 15/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9374 - mae: 1.1847 - val_loss: 1.7681 - val_mae: 1.1574
Epoch 16/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9154 - mae: 1.1918 - val_loss: 1.8051 - val_mae: 1.1644
Epoch 17/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1007 - mae: 1.2424 - val_loss: 1.8141 - val_mae: 1.1729
Epoch 18/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0609 - mae: 1.2167 - val_loss: 1.9822 - val_mae: 1.2082
Epoch 19/100
7/7 [==============================] - 0s 5ms/step - loss: 2.1135 - mae: 1.2351 - val_loss: 1.9622 - val_mae: 1.1947
Epoch 20/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0866 - mae: 1.2124 - val_loss: 2.0296 - val_mae: 1.2043
Epoch 21/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0223 - mae: 1.2191 - val_loss: 1.9198 - val_mae: 1.1934
Epoch 22/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0081 - mae: 1.2093 - val_loss: 1.8698 - val_mae: 1.1836
Epoch 23/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9646 - mae: 1.1955 - val_loss: 1.7998 - val_mae: 1.1640
Epoch 24/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9467 - mae: 1.1984 - val_loss: 1.8151 - val_mae: 1.1677
```

```
Epoch 25/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9904 - mae: 1.2108 - val_loss: 1.9197 - val_mae: 1.1885
Epoch 26/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9250 - mae: 1.1800 - val_loss: 2.3312 - val_mae: 1.2811
Epoch 27/100
7/7 [==============================] - 0s 4ms/step - loss: 2.2235 - mae: 1.2403 - val_loss: 2.0644 - val_mae: 1.2167
Epoch 28/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0348 - mae: 1.2145 - val_loss: 2.0426 - val_mae: 1.2133
Epoch 29/100
7/7 [==============================] - 0s 4ms/step - loss: 2.3500 - mae: 1.2597 - val_loss: 1.8909 - val_mae: 1.1937
Epoch 30/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1434 - mae: 1.2494 - val_loss: 2.3387 - val_mae: 1.2799
Epoch 31/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1144 - mae: 1.2026 - val_loss: 2.4896 - val_mae: 1.3133
Epoch 32/100
7/7 [==============================] - 0s 4ms/step - loss: 2.6303 - mae: 1.3503 - val_loss: 2.2369 - val_mae: 1.2515
Epoch 33/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1487 - mae: 1.2127 - val_loss: 3.1203 - val_mae: 1.4415
Epoch 34/100
7/7 [==============================] - 0s 4ms/step - loss: 2.5987 - mae: 1.3601 - val_loss: 1.8944 - val_mae: 1.1871
Epoch 35/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9279 - mae: 1.1816 - val_loss: 1.8961 - val_mae: 1.1865
Epoch 36/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0773 - mae: 1.2145 - val_loss: 2.3075 - val_mae: 1.2755
Epoch 37/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1143 - mae: 1.2166 - val_loss: 2.0257 - val_mae: 1.2129
Epoch 38/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0174 - mae: 1.1768 - val_loss: 2.0199 - val_mae: 1.2145
Epoch 39/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1483 - mae: 1.2051 - val_loss: 2.6993 - val_mae: 1.3616
Epoch 40/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1932 - mae: 1.2130 - val_loss: 2.1703 - val_mae: 1.2366
Epoch 41/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1573 - mae: 1.2190 - val_loss: 1.9490 - val_mae: 1.1983
Epoch 42/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0799 - mae: 1.2152 - val_loss: 1.9699 - val_mae: 1.2116
Epoch 43/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0056 - mae: 1.1985 - val_loss: 1.9003 - val_mae: 1.2010
Epoch 44/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0634 - mae: 1.2236 - val_loss: 2.0865 - val_mae: 1.2269
Epoch 45/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8706 - mae: 1.1553 - val_loss: 1.8948 - val_mae: 1.2003
Epoch 46/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8068 - mae: 1.1503 - val_loss: 1.8560 - val_mae: 1.1875
Epoch 47/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7935 - mae: 1.1503 - val_loss: 1.8783 - val_mae: 1.1878
Epoch 48/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8815 - mae: 1.1681 - val_loss: 1.9234 - val_mae: 1.1984
```

```
Epoch 49/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8445 - mae: 1.1532 - val_loss: 1.8528 - val_mae: 1.1836
Epoch 50/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7866 - mae: 1.1370 - val_loss: 1.9630 - val_mae: 1.2134
Epoch 51/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8439 - mae: 1.1558 - val_loss: 1.9192 - val_mae: 1.1962
Epoch 52/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7629 - mae: 1.1464 - val_loss: 1.9176 - val_mae: 1.2059
Epoch 53/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8859 - mae: 1.1673 - val_loss: 1.8489 - val_mae: 1.1851
Epoch 54/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7742 - mae: 1.1410 - val_loss: 1.8685 - val_mae: 1.1898
Epoch 55/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8024 - mae: 1.1358 - val_loss: 1.8841 - val_mae: 1.1995
Epoch 56/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7901 - mae: 1.1311 - val_loss: 1.8513 - val_mae: 1.1863
Epoch 57/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8239 - mae: 1.1556 - val_loss: 1.9292 - val_mae: 1.1994
Epoch 58/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8307 - mae: 1.1500 - val_loss: 1.8901 - val_mae: 1.2012
Epoch 59/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9379 - mae: 1.1760 - val_loss: 2.2924 - val_mae: 1.2837
Epoch 60/100
7/7 [==============================] - 0s 4ms/step - loss: 2.2630 - mae: 1.2542 - val_loss: 1.9710 - val_mae: 1.2057
Epoch 61/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8622 - mae: 1.1534 - val_loss: 1.9201 - val_mae: 1.1983
Epoch 62/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9619 - mae: 1.1833 - val_loss: 1.9226 - val_mae: 1.2094
Epoch 63/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7388 - mae: 1.1189 - val_loss: 1.8637 - val_mae: 1.1882
Epoch 64/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7648 - mae: 1.1361 - val_loss: 1.8437 - val_mae: 1.1824
Epoch 65/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7796 - mae: 1.1427 - val_loss: 1.8476 - val_mae: 1.1819
Epoch 66/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7533 - mae: 1.1321 - val_loss: 1.8826 - val_mae: 1.1914
Epoch 67/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7553 - mae: 1.1277 - val_loss: 1.8755 - val_mae: 1.1959
Epoch 68/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7928 - mae: 1.1490 - val_loss: 1.8746 - val_mae: 1.1911
Epoch 69/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8412 - mae: 1.1548 - val_loss: 1.9468 - val_mae: 1.2069
Epoch 70/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8039 - mae: 1.1406 - val_loss: 1.8936 - val_mae: 1.1967
Epoch 71/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7502 - mae: 1.1258 - val_loss: 1.9008 - val_mae: 1.2035
Epoch 72/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7934 - mae: 1.1473 - val_loss: 1.8845 - val_mae: 1.1983
```

```
Epoch 73/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7705 - mae: 1.1336 - val_loss: 1.9332 - val_mae: 1.2060
Epoch 74/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7039 - mae: 1.1188 - val_loss: 2.0437 - val_mae: 1.2271
Epoch 75/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8303 - mae: 1.1582 - val_loss: 1.8603 - val_mae: 1.1876
Epoch 76/100
7/7 [==============================] - 0s 4ms/step - loss: 1.9538 - mae: 1.1815 - val_loss: 2.1779 - val_mae: 1.2453
Epoch 77/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0263 - mae: 1.2002 - val_loss: 1.9742 - val_mae: 1.2154
Epoch 78/100
7/7 [==============================] - 0s 4ms/step - loss: 2.0589 - mae: 1.2043 - val_loss: 2.1089 - val_mae: 1.2404
Epoch 79/100
7/7 [==============================] - 0s 4ms/step - loss: 2.1557 - mae: 1.2278 - val_loss: 2.2198 - val_mae: 1.2707
Epoch 80/100
7/7 [==============================] - 0s 4ms/step - loss: 2.2199 - mae: 1.2149 - val_loss: 2.2723 - val_mae: 1.2691
Epoch 81/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8043 - mae: 1.1596 - val_loss: 1.9210 - val_mae: 1.2067
Epoch 82/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7073 - mae: 1.1047 - val_loss: 1.8644 - val_mae: 1.1888
Epoch 83/100
7/7 [==============================] - 0s 4ms/step - loss: 1.8221 - mae: 1.1480 - val_loss: 2.1590 - val_mae: 1.2490
Epoch 84/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7576 - mae: 1.1059 - val_loss: 1.9832 - val_mae: 1.2201
Epoch 85/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7491 - mae: 1.1286 - val_loss: 1.8808 - val_mae: 1.1924
Epoch 86/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7528 - mae: 1.1287 - val_loss: 1.9609 - val_mae: 1.2151
Epoch 87/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7743 - mae: 1.1350 - val_loss: 1.8944 - val_mae: 1.2007
Epoch 88/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7358 - mae: 1.1152 - val_loss: 1.9263 - val_mae: 1.2053
Epoch 89/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7031 - mae: 1.1188 - val_loss: 1.9150 - val_mae: 1.2056
Epoch 90/100
7/7 [==============================] - 0s 4ms/step - loss: 1.6864 - mae: 1.1146 - val_loss: 2.0591 - val_mae: 1.2348
Epoch 91/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7822 - mae: 1.1374 - val_loss: 1.9057 - val_mae: 1.2044
Epoch 92/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7150 - mae: 1.1177 - val_loss: 1.9110 - val_mae: 1.2057
Epoch 93/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7487 - mae: 1.1256 - val_loss: 1.9165 - val_mae: 1.2078
Epoch 94/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7102 - mae: 1.1151 - val_loss: 1.8782 - val_mae: 1.1945
Epoch 95/100
7/7 [==============================] - 0s 4ms/step - loss: 1.6976 - mae: 1.1095 - val_loss: 1.8966 - val_mae: 1.1995
Epoch 96/100
7/7 [==============================] - 0s 4ms/step - loss: 1.6986 - mae: 1.1082 - val_loss: 1.9183 - val_mae: 1.2085
```

```
Epoch 97/100
7/7 [==============================] - 0s 4ms/step - loss: 1.6817 - mae: 1.1028 - val_loss: 1.8770 - val_mae: 1.1945
Epoch 98/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7082 - mae: 1.1082 - val_loss: 1.9591 - val_mae: 1.2129
Epoch 99/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7791 - mae: 1.1194 - val_loss: 1.9299 - val_mae: 1.2028
Epoch 100/100
7/7 [==============================] - 0s 4ms/step - loss: 1.7027 - mae: 1.1165 - val_loss: 1.9008 - val_mae: 1.2022
```

Q3: Use model.summary() to display the summary of the neural network's layers and parameters

In [7]:
```
model.summary()
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense (Dense)               (None, 64)                320

 dense_1 (Dense)             (None, 32)                2080

 dense_2 (Dense)             (None, 1)                 33

=================================================================
Total params: 2433 (9.50 KB)
Trainable params: 2433 (9.50 KB)
Non-trainable params: 0 (0.00 Byte)
_________________________________________________________________
```

Q4: The number of neurons for hidden layers and hyper-parameter values should be chosen in a way that to produce the loss value of 0.05 or less for training data.

In [8]:
```
X_train_raw, X_val_raw, y_train_raw, y_val_raw = train_test_split(X, y.values.reshape(-1, 1), test_size=0.3, random_sta

# Scaling X and Y
X_scaler = MinMaxScaler()
y_scaler = MinMaxScaler()

X_scaled = X_scaler.fit_transform(X)
y_scaled = y_scaler.fit_transform(y.values.reshape(-1, 1))

X_train, X_val, y_train, y_val = train_test_split(X_scaled, y_scaled, test_size=0.3, random_state=42)
```

In [9]:
```
model_q4 = Sequential([
    Dense(256, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(128, activation='relu'),
    Dense(64, activation='relu'),
    Dense(1)
])

model_q4.compile(optimizer=Adam(learning_rate=0.0001), loss='mean_squared_error', metrics=['mae'])

history_q4 = model_q4.fit(X_train, y_train, epochs=400, validation_data=(X_val, y_val), verbose=1)
```

```
Epoch 1/400
7/7 [==============================] - 1s 21ms/step - loss: 0.3288 - mae: 0.4887 - val_loss: 0.2877 - val_mae: 0.4573
Epoch 2/400
7/7 [==============================] - 0s 5ms/step - loss: 0.2663 - mae: 0.4298 - val_loss: 0.2317 - val_mae: 0.3999
Epoch 3/400
7/7 [==============================] - 0s 5ms/step - loss: 0.2163 - mae: 0.3808 - val_loss: 0.1867 - val_mae: 0.3546
Epoch 4/400
7/7 [==============================] - 0s 5ms/step - loss: 0.1764 - mae: 0.3430 - val_loss: 0.1497 - val_mae: 0.3183
Epoch 5/400
7/7 [==============================] - 0s 5ms/step - loss: 0.1432 - mae: 0.3117 - val_loss: 0.1224 - val_mae: 0.2914
Epoch 6/400
7/7 [==============================] - 0s 5ms/step - loss: 0.1198 - mae: 0.2894 - val_loss: 0.1036 - val_mae: 0.2715
Epoch 7/400
7/7 [==============================] - 0s 6ms/step - loss: 0.1058 - mae: 0.2773 - val_loss: 0.0916 - val_mae: 0.2583
Epoch 8/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0972 - mae: 0.2690 - val_loss: 0.0852 - val_mae: 0.2503
Epoch 9/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0936 - mae: 0.2657 - val_loss: 0.0823 - val_mae: 0.2460
Epoch 10/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0927 - mae: 0.2648 - val_loss: 0.0811 - val_mae: 0.2437
Epoch 11/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0918 - mae: 0.2636 - val_loss: 0.0806 - val_mae: 0.2430
Epoch 12/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0912 - mae: 0.2624 - val_loss: 0.0802 - val_mae: 0.2430
Epoch 13/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0905 - mae: 0.2615 - val_loss: 0.0797 - val_mae: 0.2433
Epoch 14/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0898 - mae: 0.2606 - val_loss: 0.0792 - val_mae: 0.2434
Epoch 15/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0892 - mae: 0.2600 - val_loss: 0.0789 - val_mae: 0.2436
Epoch 16/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0885 - mae: 0.2589 - val_loss: 0.0785 - val_mae: 0.2432
Epoch 17/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0879 - mae: 0.2581 - val_loss: 0.0781 - val_mae: 0.2431
Epoch 18/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0873 - mae: 0.2572 - val_loss: 0.0776 - val_mae: 0.2429
Epoch 19/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0868 - mae: 0.2564 - val_loss: 0.0772 - val_mae: 0.2426
Epoch 20/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0864 - mae: 0.2558 - val_loss: 0.0770 - val_mae: 0.2427
Epoch 21/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0859 - mae: 0.2548 - val_loss: 0.0766 - val_mae: 0.2423
Epoch 22/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0853 - mae: 0.2540 - val_loss: 0.0761 - val_mae: 0.2419
Epoch 23/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0849 - mae: 0.2533 - val_loss: 0.0758 - val_mae: 0.2418
Epoch 24/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0844 - mae: 0.2525 - val_loss: 0.0756 - val_mae: 0.2418
```

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Epoch 25/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0841 - mae: 0.2524 - val_loss: 0.0755 - val_mae: 0.2421
Epoch 26/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0836 - mae: 0.2516 - val_loss: 0.0751 - val_mae: 0.2417
Epoch 27/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0832 - mae: 0.2508 - val_loss: 0.0748 - val_mae: 0.2413
Epoch 28/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0830 - mae: 0.2502 - val_loss: 0.0744 - val_mae: 0.2409
Epoch 29/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0825 - mae: 0.2494 - val_loss: 0.0742 - val_mae: 0.2406
Epoch 30/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0824 - mae: 0.2491 - val_loss: 0.0739 - val_mae: 0.2403
Epoch 31/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0820 - mae: 0.2484 - val_loss: 0.0739 - val_mae: 0.2405
Epoch 32/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0818 - mae: 0.2478 - val_loss: 0.0736 - val_mae: 0.2402
Epoch 33/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0815 - mae: 0.2473 - val_loss: 0.0737 - val_mae: 0.2403
Epoch 34/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0811 - mae: 0.2466 - val_loss: 0.0733 - val_mae: 0.2398
Epoch 35/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0809 - mae: 0.2464 - val_loss: 0.0731 - val_mae: 0.2395
Epoch 36/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0805 - mae: 0.2456 - val_loss: 0.0733 - val_mae: 0.2398
Epoch 37/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0804 - mae: 0.2454 - val_loss: 0.0735 - val_mae: 0.2404
Epoch 38/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0803 - mae: 0.2454 - val_loss: 0.0730 - val_mae: 0.2397
Epoch 39/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0800 - mae: 0.2450 - val_loss: 0.0729 - val_mae: 0.2398
Epoch 40/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0798 - mae: 0.2447 - val_loss: 0.0727 - val_mae: 0.2396
Epoch 41/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0796 - mae: 0.2443 - val_loss: 0.0723 - val_mae: 0.2389
Epoch 42/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0794 - mae: 0.2439 - val_loss: 0.0722 - val_mae: 0.2388
Epoch 43/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0793 - mae: 0.2437 - val_loss: 0.0725 - val_mae: 0.2394
Epoch 44/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0792 - mae: 0.2437 - val_loss: 0.0723 - val_mae: 0.2391
Epoch 45/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0789 - mae: 0.2430 - val_loss: 0.0716 - val_mae: 0.2380
Epoch 46/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0789 - mae: 0.2429 - val_loss: 0.0713 - val_mae: 0.2374
Epoch 47/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0787 - mae: 0.2426 - val_loss: 0.0715 - val_mae: 0.2379
Epoch 48/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0786 - mae: 0.2427 - val_loss: 0.0714 - val_mae: 0.2378
```

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Epoch 49/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0784 - mae: 0.2422 - val_loss: 0.0709 - val_mae: 0.2369
Epoch 50/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0783 - mae: 0.2417 - val_loss: 0.0708 - val_mae: 0.2366
Epoch 51/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0780 - mae: 0.2414 - val_loss: 0.0708 - val_mae: 0.2369
Epoch 52/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0781 - mae: 0.2416 - val_loss: 0.0710 - val_mae: 0.2373
Epoch 53/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0778 - mae: 0.2411 - val_loss: 0.0705 - val_mae: 0.2363
Epoch 54/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0778 - mae: 0.2409 - val_loss: 0.0703 - val_mae: 0.2359
Epoch 55/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0776 - mae: 0.2405 - val_loss: 0.0702 - val_mae: 0.2359
Epoch 56/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0775 - mae: 0.2404 - val_loss: 0.0704 - val_mae: 0.2363
Epoch 57/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0774 - mae: 0.2404 - val_loss: 0.0702 - val_mae: 0.2359
Epoch 58/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0772 - mae: 0.2400 - val_loss: 0.0699 - val_mae: 0.2353
Epoch 59/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0772 - mae: 0.2397 - val_loss: 0.0698 - val_mae: 0.2351
Epoch 60/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0770 - mae: 0.2394 - val_loss: 0.0698 - val_mae: 0.2352
Epoch 61/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0769 - mae: 0.2395 - val_loss: 0.0698 - val_mae: 0.2350
Epoch 62/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0769 - mae: 0.2397 - val_loss: 0.0698 - val_mae: 0.2352
Epoch 63/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0767 - mae: 0.2395 - val_loss: 0.0696 - val_mae: 0.2347
Epoch 64/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0769 - mae: 0.2396 - val_loss: 0.0693 - val_mae: 0.2341
Epoch 65/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0765 - mae: 0.2385 - val_loss: 0.0696 - val_mae: 0.2347
Epoch 66/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0770 - mae: 0.2399 - val_loss: 0.0702 - val_mae: 0.2359
Epoch 67/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0766 - mae: 0.2390 - val_loss: 0.0695 - val_mae: 0.2344
Epoch 68/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0763 - mae: 0.2383 - val_loss: 0.0693 - val_mae: 0.2340
Epoch 69/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0761 - mae: 0.2380 - val_loss: 0.0693 - val_mae: 0.2341
Epoch 70/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0759 - mae: 0.2378 - val_loss: 0.0694 - val_mae: 0.2344
Epoch 71/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0759 - mae: 0.2380 - val_loss: 0.0694 - val_mae: 0.2344
Epoch 72/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0758 - mae: 0.2377 - val_loss: 0.0693 - val_mae: 0.2341
```

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Epoch 73/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0757 - mae: 0.2375 - val_loss: 0.0694 - val_mae: 0.2342
Epoch 74/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0757 - mae: 0.2375 - val_loss: 0.0693 - val_mae: 0.2341
Epoch 75/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0756 - mae: 0.2371 - val_loss: 0.0692 - val_mae: 0.2339
Epoch 76/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0755 - mae: 0.2371 - val_loss: 0.0694 - val_mae: 0.2342
Epoch 77/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0754 - mae: 0.2368 - val_loss: 0.0691 - val_mae: 0.2336
Epoch 78/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0753 - mae: 0.2363 - val_loss: 0.0690 - val_mae: 0.2334
Epoch 79/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0752 - mae: 0.2362 - val_loss: 0.0691 - val_mae: 0.2336
Epoch 80/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0751 - mae: 0.2360 - val_loss: 0.0692 - val_mae: 0.2338
Epoch 81/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0751 - mae: 0.2364 - val_loss: 0.0692 - val_mae: 0.2337
Epoch 82/400
7/7 [==============================] - 0s 17ms/step - loss: 0.0750 - mae: 0.2360 - val_loss: 0.0689 - val_mae: 0.2330
Epoch 83/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0748 - mae: 0.2356 - val_loss: 0.0690 - val_mae: 0.2333
Epoch 84/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0749 - mae: 0.2356 - val_loss: 0.0690 - val_mae: 0.2333
Epoch 85/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0748 - mae: 0.2359 - val_loss: 0.0691 - val_mae: 0.2334
Epoch 86/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0747 - mae: 0.2358 - val_loss: 0.0688 - val_mae: 0.2328
Epoch 87/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0745 - mae: 0.2351 - val_loss: 0.0686 - val_mae: 0.2322
Epoch 88/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0744 - mae: 0.2347 - val_loss: 0.0685 - val_mae: 0.2320
Epoch 89/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0745 - mae: 0.2356 - val_loss: 0.0687 - val_mae: 0.2325
Epoch 90/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0742 - mae: 0.2351 - val_loss: 0.0684 - val_mae: 0.2321
Epoch 91/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0742 - mae: 0.2346 - val_loss: 0.0683 - val_mae: 0.2314
Epoch 92/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0742 - mae: 0.2341 - val_loss: 0.0683 - val_mae: 0.2316
Epoch 93/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0741 - mae: 0.2348 - val_loss: 0.0685 - val_mae: 0.2321
Epoch 94/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0741 - mae: 0.2349 - val_loss: 0.0684 - val_mae: 0.2320
Epoch 95/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0739 - mae: 0.2341 - val_loss: 0.0680 - val_mae: 0.2309
Epoch 96/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0739 - mae: 0.2337 - val_loss: 0.0681 - val_mae: 0.2310
```

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Epoch 97/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0737 - mae: 0.2336 - val_loss: 0.0680 - val_mae: 0.2311
Epoch 98/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0736 - mae: 0.2341 - val_loss: 0.0685 - val_mae: 0.2319
Epoch 99/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0738 - mae: 0.2342 - val_loss: 0.0681 - val_mae: 0.2313
Epoch 100/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0735 - mae: 0.2337 - val_loss: 0.0681 - val_mae: 0.2313
Epoch 101/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0735 - mae: 0.2335 - val_loss: 0.0679 - val_mae: 0.2308
Epoch 102/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0733 - mae: 0.2332 - val_loss: 0.0679 - val_mae: 0.2307
Epoch 103/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0732 - mae: 0.2333 - val_loss: 0.0677 - val_mae: 0.2303
Epoch 104/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0734 - mae: 0.2341 - val_loss: 0.0680 - val_mae: 0.2307
Epoch 105/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0731 - mae: 0.2333 - val_loss: 0.0676 - val_mae: 0.2300
Epoch 106/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0729 - mae: 0.2324 - val_loss: 0.0675 - val_mae: 0.2298
Epoch 107/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0729 - mae: 0.2324 - val_loss: 0.0675 - val_mae: 0.2298
Epoch 108/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0728 - mae: 0.2324 - val_loss: 0.0676 - val_mae: 0.2300
Epoch 109/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0727 - mae: 0.2320 - val_loss: 0.0675 - val_mae: 0.2296
Epoch 110/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0726 - mae: 0.2318 - val_loss: 0.0674 - val_mae: 0.2293
Epoch 111/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0726 - mae: 0.2318 - val_loss: 0.0673 - val_mae: 0.2293
Epoch 112/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0724 - mae: 0.2318 - val_loss: 0.0676 - val_mae: 0.2297
Epoch 113/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0727 - mae: 0.2328 - val_loss: 0.0680 - val_mae: 0.2305
Epoch 114/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0723 - mae: 0.2317 - val_loss: 0.0675 - val_mae: 0.2295
Epoch 115/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0723 - mae: 0.2313 - val_loss: 0.0675 - val_mae: 0.2295
Epoch 116/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0727 - mae: 0.2321 - val_loss: 0.0674 - val_mae: 0.2289
Epoch 117/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0721 - mae: 0.2309 - val_loss: 0.0673 - val_mae: 0.2291
Epoch 118/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0720 - mae: 0.2310 - val_loss: 0.0672 - val_mae: 0.2288
Epoch 119/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0720 - mae: 0.2308 - val_loss: 0.0673 - val_mae: 0.2291
Epoch 120/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0717 - mae: 0.2305 - val_loss: 0.0672 - val_mae: 0.2289
```

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Epoch 121/400
7/7 [==============================] - 0s 10ms/step - loss: 0.0717 - mae: 0.2306 - val_loss: 0.0672 - val_mae: 0.2288
Epoch 122/400
7/7 [==============================] - 0s 10ms/step - loss: 0.0717 - mae: 0.2304 - val_loss: 0.0672 - val_mae: 0.2288
Epoch 123/400
7/7 [==============================] - 0s 13ms/step - loss: 0.0716 - mae: 0.2302 - val_loss: 0.0669 - val_mae: 0.2281
Epoch 124/400
7/7 [==============================] - 0s 12ms/step - loss: 0.0714 - mae: 0.2298 - val_loss: 0.0670 - val_mae: 0.2283
Epoch 125/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0713 - mae: 0.2297 - val_loss: 0.0670 - val_mae: 0.2283
Epoch 126/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0712 - mae: 0.2294 - val_loss: 0.0668 - val_mae: 0.2279
Epoch 127/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0712 - mae: 0.2294 - val_loss: 0.0668 - val_mae: 0.2277
Epoch 128/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0710 - mae: 0.2290 - val_loss: 0.0667 - val_mae: 0.2277
Epoch 129/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0709 - mae: 0.2291 - val_loss: 0.0666 - val_mae: 0.2274
Epoch 130/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0710 - mae: 0.2295 - val_loss: 0.0664 - val_mae: 0.2271
Epoch 131/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0711 - mae: 0.2288 - val_loss: 0.0662 - val_mae: 0.2264
Epoch 132/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0708 - mae: 0.2287 - val_loss: 0.0663 - val_mae: 0.2267
Epoch 133/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0706 - mae: 0.2291 - val_loss: 0.0665 - val_mae: 0.2272
Epoch 134/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0707 - mae: 0.2290 - val_loss: 0.0664 - val_mae: 0.2270
Epoch 135/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0706 - mae: 0.2285 - val_loss: 0.0661 - val_mae: 0.2264
Epoch 136/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0704 - mae: 0.2279 - val_loss: 0.0660 - val_mae: 0.2263
Epoch 137/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0704 - mae: 0.2284 - val_loss: 0.0661 - val_mae: 0.2264
Epoch 138/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0703 - mae: 0.2283 - val_loss: 0.0663 - val_mae: 0.2267
Epoch 139/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0709 - mae: 0.2288 - val_loss: 0.0660 - val_mae: 0.2259
Epoch 140/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0701 - mae: 0.2277 - val_loss: 0.0661 - val_mae: 0.2262
Epoch 141/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0700 - mae: 0.2281 - val_loss: 0.0663 - val_mae: 0.2265
Epoch 142/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0699 - mae: 0.2272 - val_loss: 0.0662 - val_mae: 0.2260
Epoch 143/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0700 - mae: 0.2271 - val_loss: 0.0659 - val_mae: 0.2255
Epoch 144/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0698 - mae: 0.2268 - val_loss: 0.0661 - val_mae: 0.2259
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Epoch 145/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0699 - mae: 0.2278 - val_loss: 0.0661 - val_mae: 0.2259
Epoch 146/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0694 - mae: 0.2263 - val_loss: 0.0657 - val_mae: 0.2250
Epoch 147/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0698 - mae: 0.2271 - val_loss: 0.0657 - val_mae: 0.2252
Epoch 148/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0694 - mae: 0.2267 - val_loss: 0.0659 - val_mae: 0.2255
Epoch 149/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0693 - mae: 0.2264 - val_loss: 0.0656 - val_mae: 0.2250
Epoch 150/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0694 - mae: 0.2262 - val_loss: 0.0654 - val_mae: 0.2241
Epoch 151/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0692 - mae: 0.2258 - val_loss: 0.0656 - val_mae: 0.2249
Epoch 152/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0690 - mae: 0.2260 - val_loss: 0.0658 - val_mae: 0.2251
Epoch 153/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0691 - mae: 0.2261 - val_loss: 0.0656 - val_mae: 0.2248
Epoch 154/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0688 - mae: 0.2256 - val_loss: 0.0657 - val_mae: 0.2249
Epoch 155/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0687 - mae: 0.2253 - val_loss: 0.0657 - val_mae: 0.2249
Epoch 156/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0686 - mae: 0.2250 - val_loss: 0.0656 - val_mae: 0.2247
Epoch 157/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0686 - mae: 0.2251 - val_loss: 0.0653 - val_mae: 0.2241
Epoch 158/400
7/7 [==============================] - 0s 13ms/step - loss: 0.0684 - mae: 0.2250 - val_loss: 0.0655 - val_mae: 0.2245
Epoch 159/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0684 - mae: 0.2247 - val_loss: 0.0651 - val_mae: 0.2238
Epoch 160/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0683 - mae: 0.2248 - val_loss: 0.0648 - val_mae: 0.2232
Epoch 161/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0684 - mae: 0.2249 - val_loss: 0.0649 - val_mae: 0.2234
Epoch 162/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0682 - mae: 0.2248 - val_loss: 0.0648 - val_mae: 0.2231
Epoch 163/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0680 - mae: 0.2241 - val_loss: 0.0649 - val_mae: 0.2231
Epoch 164/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0680 - mae: 0.2237 - val_loss: 0.0652 - val_mae: 0.2237
Epoch 165/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0679 - mae: 0.2241 - val_loss: 0.0650 - val_mae: 0.2233
Epoch 166/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0678 - mae: 0.2239 - val_loss: 0.0650 - val_mae: 0.2232
Epoch 167/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0678 - mae: 0.2234 - val_loss: 0.0646 - val_mae: 0.2221
Epoch 168/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0679 - mae: 0.2238 - val_loss: 0.0649 - val_mae: 0.2229
```

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Epoch 169/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0674 - mae: 0.2226 - val_loss: 0.0646 - val_mae: 0.2223
Epoch 170/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0674 - mae: 0.2230 - val_loss: 0.0644 - val_mae: 0.2221
Epoch 171/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0679 - mae: 0.2236 - val_loss: 0.0642 - val_mae: 0.2216
Epoch 172/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0672 - mae: 0.2224 - val_loss: 0.0649 - val_mae: 0.2226
Epoch 173/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0674 - mae: 0.2236 - val_loss: 0.0648 - val_mae: 0.2226
Epoch 174/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0675 - mae: 0.2230 - val_loss: 0.0645 - val_mae: 0.2218
Epoch 175/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0678 - mae: 0.2244 - val_loss: 0.0650 - val_mae: 0.2227
Epoch 176/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0675 - mae: 0.2230 - val_loss: 0.0640 - val_mae: 0.2205
Epoch 177/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0667 - mae: 0.2216 - val_loss: 0.0640 - val_mae: 0.2209
Epoch 178/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0666 - mae: 0.2218 - val_loss: 0.0643 - val_mae: 0.2212
Epoch 179/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0665 - mae: 0.2215 - val_loss: 0.0640 - val_mae: 0.2208
Epoch 180/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0665 - mae: 0.2213 - val_loss: 0.0639 - val_mae: 0.2205
Epoch 181/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0669 - mae: 0.2214 - val_loss: 0.0639 - val_mae: 0.2196
Epoch 182/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0664 - mae: 0.2209 - val_loss: 0.0639 - val_mae: 0.2205
Epoch 183/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0662 - mae: 0.2210 - val_loss: 0.0641 - val_mae: 0.2208
Epoch 184/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0660 - mae: 0.2206 - val_loss: 0.0636 - val_mae: 0.2200
Epoch 185/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0661 - mae: 0.2200 - val_loss: 0.0635 - val_mae: 0.2195
Epoch 186/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0661 - mae: 0.2205 - val_loss: 0.0640 - val_mae: 0.2205
Epoch 187/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0659 - mae: 0.2202 - val_loss: 0.0637 - val_mae: 0.2198
Epoch 188/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0657 - mae: 0.2197 - val_loss: 0.0638 - val_mae: 0.2201
Epoch 189/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0661 - mae: 0.2206 - val_loss: 0.0641 - val_mae: 0.2206
Epoch 190/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0657 - mae: 0.2200 - val_loss: 0.0639 - val_mae: 0.2202
Epoch 191/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0653 - mae: 0.2190 - val_loss: 0.0635 - val_mae: 0.2193
Epoch 192/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0653 - mae: 0.2191 - val_loss: 0.0634 - val_mae: 0.2192
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Epoch 193/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0653 - mae: 0.2197 - val_loss: 0.0636 - val_mae: 0.2194
Epoch 194/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0650 - mae: 0.2189 - val_loss: 0.0631 - val_mae: 0.2184
Epoch 195/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0652 - mae: 0.2187 - val_loss: 0.0630 - val_mae: 0.2181
Epoch 196/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0656 - mae: 0.2191 - val_loss: 0.0635 - val_mae: 0.2192
Epoch 197/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0647 - mae: 0.2183 - val_loss: 0.0633 - val_mae: 0.2187
Epoch 198/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0652 - mae: 0.2189 - val_loss: 0.0633 - val_mae: 0.2176
Epoch 199/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0658 - mae: 0.2202 - val_loss: 0.0637 - val_mae: 0.2193
Epoch 200/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0646 - mae: 0.2179 - val_loss: 0.0635 - val_mae: 0.2191
Epoch 201/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0643 - mae: 0.2171 - val_loss: 0.0631 - val_mae: 0.2174
Epoch 202/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0648 - mae: 0.2178 - val_loss: 0.0628 - val_mae: 0.2174
Epoch 203/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0642 - mae: 0.2173 - val_loss: 0.0630 - val_mae: 0.2180
Epoch 204/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0645 - mae: 0.2176 - val_loss: 0.0634 - val_mae: 0.2188
Epoch 205/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0642 - mae: 0.2173 - val_loss: 0.0624 - val_mae: 0.2166
Epoch 206/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0642 - mae: 0.2171 - val_loss: 0.0622 - val_mae: 0.2162
Epoch 207/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0641 - mae: 0.2174 - val_loss: 0.0627 - val_mae: 0.2174
Epoch 208/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0638 - mae: 0.2165 - val_loss: 0.0623 - val_mae: 0.2164
Epoch 209/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0644 - mae: 0.2172 - val_loss: 0.0624 - val_mae: 0.2161
Epoch 210/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0634 - mae: 0.2153 - val_loss: 0.0631 - val_mae: 0.2182
Epoch 211/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0638 - mae: 0.2164 - val_loss: 0.0624 - val_mae: 0.2166
Epoch 212/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0633 - mae: 0.2154 - val_loss: 0.0621 - val_mae: 0.2159
Epoch 213/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0635 - mae: 0.2156 - val_loss: 0.0621 - val_mae: 0.2160
Epoch 214/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0631 - mae: 0.2152 - val_loss: 0.0624 - val_mae: 0.2167
Epoch 215/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0630 - mae: 0.2150 - val_loss: 0.0627 - val_mae: 0.2172
Epoch 216/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0630 - mae: 0.2148 - val_loss: 0.0624 - val_mae: 0.2164
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Epoch 217/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0629 - mae: 0.2146 - val_loss: 0.0616 - val_mae: 0.2145
Epoch 218/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0628 - mae: 0.2145 - val_loss: 0.0617 - val_mae: 0.2148
Epoch 219/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0628 - mae: 0.2145 - val_loss: 0.0619 - val_mae: 0.2154
Epoch 220/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0630 - mae: 0.2147 - val_loss: 0.0620 - val_mae: 0.2146
Epoch 221/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0628 - mae: 0.2145 - val_loss: 0.0620 - val_mae: 0.2155
Epoch 222/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0631 - mae: 0.2147 - val_loss: 0.0626 - val_mae: 0.2170
Epoch 223/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0627 - mae: 0.2142 - val_loss: 0.0617 - val_mae: 0.2139
Epoch 224/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0623 - mae: 0.2134 - val_loss: 0.0618 - val_mae: 0.2149
Epoch 225/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0619 - mae: 0.2132 - val_loss: 0.0616 - val_mae: 0.2149
Epoch 226/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0624 - mae: 0.2138 - val_loss: 0.0616 - val_mae: 0.2148
Epoch 227/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0615 - mae: 0.2118 - val_loss: 0.0612 - val_mae: 0.2129
Epoch 228/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0619 - mae: 0.2126 - val_loss: 0.0612 - val_mae: 0.2134
Epoch 229/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0617 - mae: 0.2119 - val_loss: 0.0616 - val_mae: 0.2144
Epoch 230/400
7/7 [==============================] - 0s 15ms/step - loss: 0.0614 - mae: 0.2116 - val_loss: 0.0617 - val_mae: 0.2145
Epoch 231/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0615 - mae: 0.2119 - val_loss: 0.0612 - val_mae: 0.2135
Epoch 232/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0615 - mae: 0.2117 - val_loss: 0.0613 - val_mae: 0.2139
Epoch 233/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0612 - mae: 0.2112 - val_loss: 0.0608 - val_mae: 0.2128
Epoch 234/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0610 - mae: 0.2109 - val_loss: 0.0609 - val_mae: 0.2125
Epoch 235/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0609 - mae: 0.2109 - val_loss: 0.0613 - val_mae: 0.2136
Epoch 236/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0613 - mae: 0.2113 - val_loss: 0.0614 - val_mae: 0.2140
Epoch 237/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0605 - mae: 0.2103 - val_loss: 0.0604 - val_mae: 0.2115
Epoch 238/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0606 - mae: 0.2103 - val_loss: 0.0604 - val_mae: 0.2117
Epoch 239/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0603 - mae: 0.2097 - val_loss: 0.0605 - val_mae: 0.2124
Epoch 240/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0608 - mae: 0.2103 - val_loss: 0.0612 - val_mae: 0.2141
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Epoch 241/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0603 - mae: 0.2101 - val_loss: 0.0605 - val_mae: 0.2107
Epoch 242/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0611 - mae: 0.2113 - val_loss: 0.0606 - val_mae: 0.2129
Epoch 243/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0603 - mae: 0.2098 - val_loss: 0.0601 - val_mae: 0.2112
Epoch 244/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0599 - mae: 0.2089 - val_loss: 0.0602 - val_mae: 0.2110
Epoch 245/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0598 - mae: 0.2086 - val_loss: 0.0605 - val_mae: 0.2115
Epoch 246/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0597 - mae: 0.2082 - val_loss: 0.0605 - val_mae: 0.2120
Epoch 247/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0595 - mae: 0.2081 - val_loss: 0.0602 - val_mae: 0.2110
Epoch 248/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0595 - mae: 0.2081 - val_loss: 0.0599 - val_mae: 0.2105
Epoch 249/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0593 - mae: 0.2076 - val_loss: 0.0599 - val_mae: 0.2103
Epoch 250/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0594 - mae: 0.2080 - val_loss: 0.0604 - val_mae: 0.2115
Epoch 251/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0590 - mae: 0.2070 - val_loss: 0.0599 - val_mae: 0.2108
Epoch 252/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0591 - mae: 0.2070 - val_loss: 0.0601 - val_mae: 0.2108
Epoch 253/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0589 - mae: 0.2065 - val_loss: 0.0608 - val_mae: 0.2133
Epoch 254/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0588 - mae: 0.2066 - val_loss: 0.0596 - val_mae: 0.2098
Epoch 255/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0584 - mae: 0.2060 - val_loss: 0.0596 - val_mae: 0.2092
Epoch 256/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0587 - mae: 0.2062 - val_loss: 0.0598 - val_mae: 0.2098
Epoch 257/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0584 - mae: 0.2058 - val_loss: 0.0599 - val_mae: 0.2111
Epoch 258/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0586 - mae: 0.2065 - val_loss: 0.0601 - val_mae: 0.2118
Epoch 259/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0581 - mae: 0.2050 - val_loss: 0.0589 - val_mae: 0.2077
Epoch 260/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0583 - mae: 0.2057 - val_loss: 0.0590 - val_mae: 0.2078
Epoch 261/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0583 - mae: 0.2051 - val_loss: 0.0599 - val_mae: 0.2119
Epoch 262/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0579 - mae: 0.2045 - val_loss: 0.0590 - val_mae: 0.2088
Epoch 263/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0580 - mae: 0.2051 - val_loss: 0.0590 - val_mae: 0.2073
Epoch 264/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0577 - mae: 0.2045 - val_loss: 0.0593 - val_mae: 0.2100
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Epoch 265/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0576 - mae: 0.2040 - val_loss: 0.0593 - val_mae: 0.2102
Epoch 266/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0573 - mae: 0.2039 - val_loss: 0.0590 - val_mae: 0.2079
Epoch 267/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0574 - mae: 0.2039 - val_loss: 0.0590 - val_mae: 0.2086
Epoch 268/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0571 - mae: 0.2034 - val_loss: 0.0590 - val_mae: 0.2086
Epoch 269/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0574 - mae: 0.2034 - val_loss: 0.0595 - val_mae: 0.2107
Epoch 270/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0569 - mae: 0.2021 - val_loss: 0.0584 - val_mae: 0.2065
Epoch 271/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0569 - mae: 0.2033 - val_loss: 0.0583 - val_mae: 0.2077
Epoch 272/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0568 - mae: 0.2023 - val_loss: 0.0586 - val_mae: 0.2085
Epoch 273/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0571 - mae: 0.2031 - val_loss: 0.0583 - val_mae: 0.2062
Epoch 274/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0567 - mae: 0.2025 - val_loss: 0.0588 - val_mae: 0.2092
Epoch 275/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0564 - mae: 0.2018 - val_loss: 0.0586 - val_mae: 0.2087
Epoch 276/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0562 - mae: 0.2013 - val_loss: 0.0580 - val_mae: 0.2069
Epoch 277/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0563 - mae: 0.2014 - val_loss: 0.0579 - val_mae: 0.2071
Epoch 278/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0564 - mae: 0.2016 - val_loss: 0.0583 - val_mae: 0.2091
Epoch 279/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0559 - mae: 0.2005 - val_loss: 0.0577 - val_mae: 0.2064
Epoch 280/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0561 - mae: 0.2011 - val_loss: 0.0579 - val_mae: 0.2057
Epoch 281/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0556 - mae: 0.2002 - val_loss: 0.0589 - val_mae: 0.2107
Epoch 282/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0561 - mae: 0.2004 - val_loss: 0.0576 - val_mae: 0.2071
Epoch 283/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0558 - mae: 0.2007 - val_loss: 0.0572 - val_mae: 0.2044
Epoch 284/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0553 - mae: 0.1992 - val_loss: 0.0580 - val_mae: 0.2080
Epoch 285/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0554 - mae: 0.1994 - val_loss: 0.0575 - val_mae: 0.2052
Epoch 286/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0548 - mae: 0.1985 - val_loss: 0.0576 - val_mae: 0.2071
Epoch 287/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0552 - mae: 0.1992 - val_loss: 0.0573 - val_mae: 0.2068
Epoch 288/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0547 - mae: 0.1992 - val_loss: 0.0571 - val_mae: 0.2044
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Epoch 289/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0546 - mae: 0.1978 - val_loss: 0.0574 - val_mae: 0.2070
Epoch 290/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0549 - mae: 0.1985 - val_loss: 0.0577 - val_mae: 0.2072
Epoch 291/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0543 - mae: 0.1972 - val_loss: 0.0578 - val_mae: 0.2069
Epoch 292/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0540 - mae: 0.1966 - val_loss: 0.0573 - val_mae: 0.2049
Epoch 293/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0541 - mae: 0.1970 - val_loss: 0.0572 - val_mae: 0.2053
Epoch 294/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0542 - mae: 0.1977 - val_loss: 0.0571 - val_mae: 0.2064
Epoch 295/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0541 - mae: 0.1975 - val_loss: 0.0569 - val_mae: 0.2049
Epoch 296/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0537 - mae: 0.1968 - val_loss: 0.0572 - val_mae: 0.2062
Epoch 297/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0536 - mae: 0.1960 - val_loss: 0.0569 - val_mae: 0.2051
Epoch 298/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0538 - mae: 0.1966 - val_loss: 0.0561 - val_mae: 0.2026
Epoch 299/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0533 - mae: 0.1951 - val_loss: 0.0569 - val_mae: 0.2064
Epoch 300/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0534 - mae: 0.1957 - val_loss: 0.0564 - val_mae: 0.2044
Epoch 301/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0530 - mae: 0.1950 - val_loss: 0.0561 - val_mae: 0.2028
Epoch 302/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0532 - mae: 0.1960 - val_loss: 0.0560 - val_mae: 0.2024
Epoch 303/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0530 - mae: 0.1956 - val_loss: 0.0562 - val_mae: 0.2039
Epoch 304/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0528 - mae: 0.1947 - val_loss: 0.0573 - val_mae: 0.2079
Epoch 305/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0529 - mae: 0.1946 - val_loss: 0.0562 - val_mae: 0.2039
Epoch 306/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0523 - mae: 0.1937 - val_loss: 0.0560 - val_mae: 0.2025
Epoch 307/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0528 - mae: 0.1946 - val_loss: 0.0562 - val_mae: 0.2030
Epoch 308/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0529 - mae: 0.1943 - val_loss: 0.0567 - val_mae: 0.2060
Epoch 309/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0519 - mae: 0.1929 - val_loss: 0.0558 - val_mae: 0.2018
Epoch 310/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0525 - mae: 0.1936 - val_loss: 0.0559 - val_mae: 0.2027
Epoch 311/400
7/7 [==============================] - 0s 10ms/step - loss: 0.0524 - mae: 0.1937 - val_loss: 0.0560 - val_mae: 0.2042
Epoch 312/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0520 - mae: 0.1931 - val_loss: 0.0549 - val_mae: 0.2002
```

```
Epoch 313/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0516 - mae: 0.1928 - val_loss: 0.0551 - val_mae: 0.2022
Epoch 314/400
7/7 [==============================] - 0s 22ms/step - loss: 0.0518 - mae: 0.1931 - val_loss: 0.0550 - val_mae: 0.2011
Epoch 315/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0515 - mae: 0.1922 - val_loss: 0.0551 - val_mae: 0.2016
Epoch 316/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0524 - mae: 0.1937 - val_loss: 0.0550 - val_mae: 0.2003
Epoch 317/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0511 - mae: 0.1901 - val_loss: 0.0567 - val_mae: 0.2066
Epoch 318/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0516 - mae: 0.1914 - val_loss: 0.0555 - val_mae: 0.2028
Epoch 319/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0514 - mae: 0.1917 - val_loss: 0.0547 - val_mae: 0.2000
Epoch 320/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0507 - mae: 0.1908 - val_loss: 0.0551 - val_mae: 0.2025
Epoch 321/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0509 - mae: 0.1910 - val_loss: 0.0549 - val_mae: 0.2018
Epoch 322/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0509 - mae: 0.1905 - val_loss: 0.0551 - val_mae: 0.2025
Epoch 323/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0510 - mae: 0.1898 - val_loss: 0.0542 - val_mae: 0.1982
Epoch 324/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0505 - mae: 0.1891 - val_loss: 0.0550 - val_mae: 0.2025
Epoch 325/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0506 - mae: 0.1902 - val_loss: 0.0545 - val_mae: 0.2001
Epoch 326/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0503 - mae: 0.1894 - val_loss: 0.0545 - val_mae: 0.2005
Epoch 327/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0507 - mae: 0.1897 - val_loss: 0.0546 - val_mae: 0.2015
Epoch 328/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0500 - mae: 0.1885 - val_loss: 0.0538 - val_mae: 0.1982
Epoch 329/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0500 - mae: 0.1883 - val_loss: 0.0538 - val_mae: 0.1983
Epoch 330/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0499 - mae: 0.1881 - val_loss: 0.0543 - val_mae: 0.2006
Epoch 331/400
7/7 [==============================] - 0s 4ms/step - loss: 0.0497 - mae: 0.1878 - val_loss: 0.0542 - val_mae: 0.1990
Epoch 332/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0497 - mae: 0.1879 - val_loss: 0.0541 - val_mae: 0.1997
Epoch 333/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0494 - mae: 0.1874 - val_loss: 0.0541 - val_mae: 0.1999
Epoch 334/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0496 - mae: 0.1876 - val_loss: 0.0541 - val_mae: 0.2005
Epoch 335/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0493 - mae: 0.1872 - val_loss: 0.0532 - val_mae: 0.1966
Epoch 336/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0494 - mae: 0.1872 - val_loss: 0.0540 - val_mae: 0.2004
```

```
Epoch 337/400
7/7 [==============================] - 0s 27ms/step - loss: 0.0491 - mae: 0.1865 - val_loss: 0.0535 - val_mae: 0.1984
Epoch 338/400
7/7 [==============================] - 0s 13ms/step - loss: 0.0492 - mae: 0.1863 - val_loss: 0.0536 - val_mae: 0.1984
Epoch 339/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0486 - mae: 0.1856 - val_loss: 0.0532 - val_mae: 0.1969
Epoch 340/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0485 - mae: 0.1856 - val_loss: 0.0536 - val_mae: 0.1989
Epoch 341/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0486 - mae: 0.1852 - val_loss: 0.0537 - val_mae: 0.1989
Epoch 342/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0485 - mae: 0.1846 - val_loss: 0.0544 - val_mae: 0.2011
Epoch 343/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0481 - mae: 0.1840 - val_loss: 0.0529 - val_mae: 0.1962
Epoch 344/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0488 - mae: 0.1858 - val_loss: 0.0525 - val_mae: 0.1953
Epoch 345/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0481 - mae: 0.1843 - val_loss: 0.0533 - val_mae: 0.1983
Epoch 346/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0482 - mae: 0.1840 - val_loss: 0.0527 - val_mae: 0.1962
Epoch 347/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0478 - mae: 0.1834 - val_loss: 0.0534 - val_mae: 0.1991
Epoch 348/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0482 - mae: 0.1841 - val_loss: 0.0530 - val_mae: 0.1971
Epoch 349/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0487 - mae: 0.1847 - val_loss: 0.0523 - val_mae: 0.1945
Epoch 350/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0474 - mae: 0.1829 - val_loss: 0.0536 - val_mae: 0.1995
Epoch 351/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0477 - mae: 0.1837 - val_loss: 0.0529 - val_mae: 0.1969
Epoch 352/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0472 - mae: 0.1825 - val_loss: 0.0528 - val_mae: 0.1971
Epoch 353/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0476 - mae: 0.1824 - val_loss: 0.0528 - val_mae: 0.1971
Epoch 354/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0469 - mae: 0.1816 - val_loss: 0.0521 - val_mae: 0.1945
Epoch 355/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0472 - mae: 0.1819 - val_loss: 0.0519 - val_mae: 0.1944
Epoch 356/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0469 - mae: 0.1815 - val_loss: 0.0525 - val_mae: 0.1967
Epoch 357/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0467 - mae: 0.1810 - val_loss: 0.0523 - val_mae: 0.1950
Epoch 358/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0470 - mae: 0.1812 - val_loss: 0.0533 - val_mae: 0.1983
Epoch 359/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0469 - mae: 0.1811 - val_loss: 0.0525 - val_mae: 0.1965
Epoch 360/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0471 - mae: 0.1816 - val_loss: 0.0523 - val_mae: 0.1955
```

```
Epoch 361/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0480 - mae: 0.1803 - val_loss: 0.0517 - val_mae: 0.1925
Epoch 362/400
7/7 [==============================] - 0s 14ms/step - loss: 0.0466 - mae: 0.1784 - val_loss: 0.0538 - val_mae: 0.1997
Epoch 363/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0473 - mae: 0.1827 - val_loss: 0.0514 - val_mae: 0.1930
Epoch 364/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0459 - mae: 0.1794 - val_loss: 0.0520 - val_mae: 0.1952
Epoch 365/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0462 - mae: 0.1795 - val_loss: 0.0514 - val_mae: 0.1933
Epoch 366/400
7/7 [==============================] - 0s 10ms/step - loss: 0.0462 - mae: 0.1794 - val_loss: 0.0514 - val_mae: 0.1934
Epoch 367/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0455 - mae: 0.1783 - val_loss: 0.0523 - val_mae: 0.1961
Epoch 368/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0457 - mae: 0.1781 - val_loss: 0.0513 - val_mae: 0.1933
Epoch 369/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0457 - mae: 0.1783 - val_loss: 0.0514 - val_mae: 0.1935
Epoch 370/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0455 - mae: 0.1779 - val_loss: 0.0529 - val_mae: 0.1974
Epoch 371/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0456 - mae: 0.1785 - val_loss: 0.0505 - val_mae: 0.1909
Epoch 372/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0453 - mae: 0.1782 - val_loss: 0.0512 - val_mae: 0.1932
Epoch 373/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0455 - mae: 0.1776 - val_loss: 0.0521 - val_mae: 0.1954
Epoch 374/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0452 - mae: 0.1768 - val_loss: 0.0505 - val_mae: 0.1914
Epoch 375/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0449 - mae: 0.1763 - val_loss: 0.0522 - val_mae: 0.1954
Epoch 376/400
7/7 [==============================] - 0s 5ms/step - loss: 0.0450 - mae: 0.1764 - val_loss: 0.0512 - val_mae: 0.1930
Epoch 377/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0445 - mae: 0.1759 - val_loss: 0.0506 - val_mae: 0.1916
Epoch 378/400
7/7 [==============================] - 0s 11ms/step - loss: 0.0453 - mae: 0.1769 - val_loss: 0.0511 - val_mae: 0.1930
Epoch 379/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0443 - mae: 0.1752 - val_loss: 0.0511 - val_mae: 0.1930
Epoch 380/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0446 - mae: 0.1754 - val_loss: 0.0507 - val_mae: 0.1922
Epoch 381/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0441 - mae: 0.1751 - val_loss: 0.0505 - val_mae: 0.1914
Epoch 382/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0443 - mae: 0.1753 - val_loss: 0.0506 - val_mae: 0.1919
Epoch 383/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0440 - mae: 0.1745 - val_loss: 0.0504 - val_mae: 0.1911
Epoch 384/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0440 - mae: 0.1743 - val_loss: 0.0520 - val_mae: 0.1947
```

```
Epoch 385/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0446 - mae: 0.1740 - val_loss: 0.0502 - val_mae: 0.1908
Epoch 386/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0445 - mae: 0.1743 - val_loss: 0.0499 - val_mae: 0.1901
Epoch 387/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0437 - mae: 0.1731 - val_loss: 0.0511 - val_mae: 0.1928
Epoch 388/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0440 - mae: 0.1732 - val_loss: 0.0501 - val_mae: 0.1907
Epoch 389/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0435 - mae: 0.1731 - val_loss: 0.0499 - val_mae: 0.1905
Epoch 390/400
7/7 [==============================] - 0s 9ms/step - loss: 0.0434 - mae: 0.1729 - val_loss: 0.0501 - val_mae: 0.1908
Epoch 391/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0439 - mae: 0.1722 - val_loss: 0.0501 - val_mae: 0.1904
Epoch 392/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0436 - mae: 0.1732 - val_loss: 0.0500 - val_mae: 0.1907
Epoch 393/400
7/7 [==============================] - 0s 22ms/step - loss: 0.0441 - mae: 0.1740 - val_loss: 0.0502 - val_mae: 0.1905
Epoch 394/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0430 - mae: 0.1720 - val_loss: 0.0496 - val_mae: 0.1895
Epoch 395/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0431 - mae: 0.1720 - val_loss: 0.0505 - val_mae: 0.1918
Epoch 396/400
7/7 [==============================] - 0s 6ms/step - loss: 0.0432 - mae: 0.1725 - val_loss: 0.0501 - val_mae: 0.1908
Epoch 397/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0429 - mae: 0.1717 - val_loss: 0.0498 - val_mae: 0.1900
Epoch 398/400
7/7 [==============================] - 0s 8ms/step - loss: 0.0428 - mae: 0.1715 - val_loss: 0.0502 - val_mae: 0.1905
Epoch 399/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0426 - mae: 0.1706 - val_loss: 0.0491 - val_mae: 0.1884
Epoch 400/400
7/7 [==============================] - 0s 7ms/step - loss: 0.0429 - mae: 0.1711 - val_loss: 0.0502 - val_mae: 0.1907
```

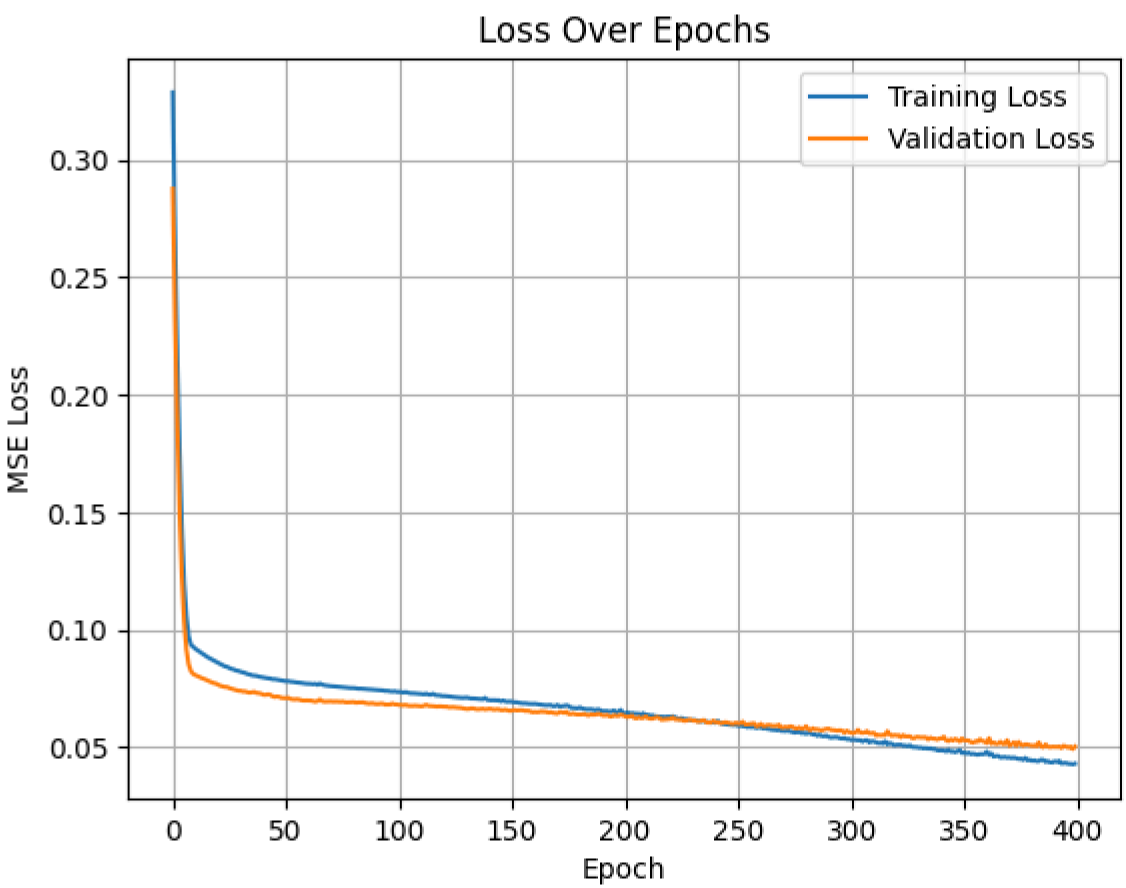
In [10]:
```python
print("Final training loss (Q4):", history_q4.history['loss'][-1])
```

```
Final training loss (Q4): 0.04291306063532829
```

Q5: Plot the loss values of all epochs for both training and validation data.

In [11]:
```python
plt.plot(history_q4.history['loss'], label='Training Loss')
plt.plot(history_q4.history['val_loss'], label='Validation Loss')
plt.title('Loss Over Epochs')
plt.xlabel('Epoch')
plt.ylabel('MSE Loss')
plt.legend()
```

```
plt.grid(True)
plt.show()
```



Q6: Plot the predicted values vs the target values for validation data

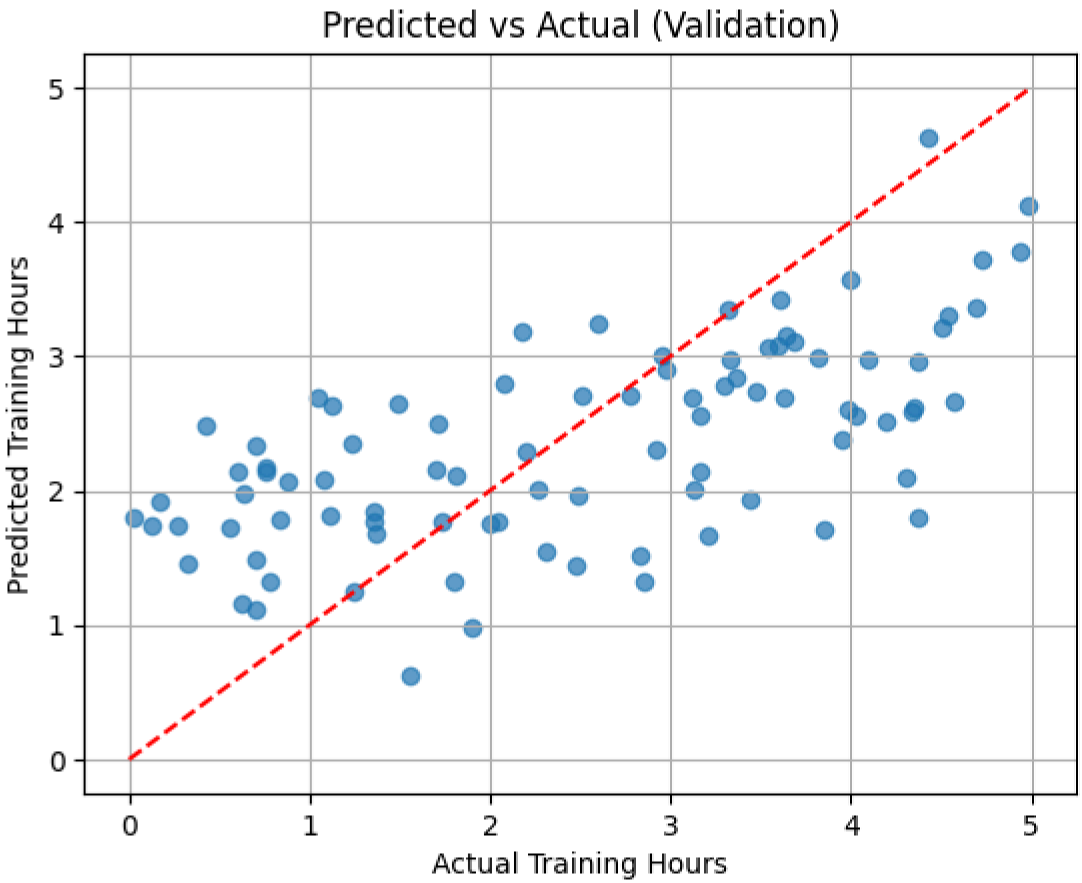
In [12]:
```
# predict
y_val_pred_scaled = model_q4.predict(X_val)

# unscale pred and target
y_val_pred = y_scaler.inverse_transform(y_val_pred_scaled)
```

```
3/3 [==============================] - 0s 1ms/step
```

In [13]:
```
plt.scatter(y_val_raw, y_val_pred, alpha=0.7)
plt.plot([0, 5], [0, 5], 'r--')
plt.xlabel('Actual Training Hours')
plt.ylabel('Predicted Training Hours')
plt.title('Predicted vs Actual (Validation)')
```

```
plt.grid(True)
plt.show()
```



In [ ]: